Show Code

Diabetes Risk Analysis & Public Health Insights

US BRFSS 2023 • SQL • Clustering • Streamlit Dashboard • Latin America Context

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Objective: Analyze diabetes risk factors using BRFSS 2023 data (USA), build predictive models, perform clustering for population segmentation, and contextualize findings with Latin America trends and projections (PAHO & IDF data).

Project Overview

This project aims to:

- Perform exploratory data analysis (EDA) on diabetes indicators from the 2023 Behavioral Risk Factor Surveillance System (BRFSS).
- Identify key risk factors and correlations.
- Build a classification model to predict diabetes/pre-diabetes risk.
- Apply clustering to segment populations by risk profiles.
- Enrich the analysis with regional data from Latin America (prevalence trends, burden, mortality, treatment gaps).
- Provide actionable public health insights and recommendations, especially relevant for Argentina and Latin America.

Datasets

1. **Main Dataset**
   - Source: CDC BRFSS 2023 (cleaned version from Kaggle)
   - File: `diabetes_012_health_indicators_BRFSS2023.csv`
   - Rows: ~261,589
   - Target: Diabetes_012 (0 = no diabetes, 1 = pre-diabetes, 2 = diabetes)
   - Key features: BMI, HighBP, GenHlth, PhysActivity, Age, Income, etc.
2. **Latam Context Datasets**
   - IDF Diabetes Atlas – South & Central America projections (2000–2050)
     File: `idf_south_central_america_formatted.csv`
   - PAHO Prevalence and Treatment Coverage (1990–2022)
     File: `paho_prevalence_trends.csv`

- PAHO Burden of Diabetes 2019 (by subregion)
  File: `paho_burden_2019.csv`
- PAHO Deaths by Country (2019)
  File: `paho_level_by_country.csv`
- PAHO Summary – Treatment Gap (1990–2022)
  File: `paho_diabetes_in_the_americas_summary_of_estimates.csv`

Objectives

Main Objectives

- Understand diabetes prevalence and risk factors in the US population (2023 data).
- Build and evaluate a predictive model for diabetes risk classification.
- Identify population segments with similar risk profiles using clustering.
- Compare US patterns with Latin America trends and future projections.

Technical Goals

- Demonstrate advanced SQL usage (DuckDB) for data preparation and aggregation.
- Perform complete EDA with interactive visualizations (Plotly).
- Apply supervised (classification) and unsupervised (clustering) machine learning.
- Build an interactive dashboard (Streamlit) for exploration and insights.

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1. Data Preparation & Cleaning

1.1. Data Preparation

```
Table 'cdc' loaded.
Table 'idf' loaded.
Table 'paho_prevalence' loaded.
Table 'paho_burden' loaded.
Table 'paho_deaths_country' loaded.
Table 'paho_summary' loaded.

=== CDC ===
Rows: 261,589

Columns:
             column_name column_type
0           Diabetes_012      DOUBLE
1          KidneyDisease      DOUBLE
2                 HighBP      DOUBLE
3               HighChol      DOUBLE
4              CholCheck      DOUBLE
5                 Asthma      DOUBLE
6                   COPD      DOUBLE
7                    BMI      DOUBLE
8                 Smoker      DOUBLE
9                 Stroke      DOUBLE
10  HeartDiseaseorAttack      DOUBLE
11          PhysActivity      DOUBLE
12     HvyAlcoholConsump      DOUBLE
13         AnyHealthcare      DOUBLE
14           NoDocbcCost      DOUBLE
15               GenHlth      DOUBLE
16              MentHlth      DOUBLE
17              PhysHlth      DOUBLE
18              DiffWalk      DOUBLE
19                   Sex      DOUBLE
20              AgeGroup      DOUBLE
21             Education      DOUBLE
22                Income      DOUBLE

First 3 rows:
   Diabetes_012  KidneyDisease  HighBP  HighChol  CholCheck  Asthma  COPD  \
0           0.0            2.0     1.0       1.0        1.0     1.0   2.0
1           2.0            2.0     1.0       0.0        1.0     2.0   2.0
2           0.0            2.0     1.0       1.0        1.0     2.0   2.0

    BMI  Smoker  Stroke  HeartDiseaseorAttack  PhysActivity  \
0  22.0     1.0     0.0                   0.0           1.0
1  26.0     0.0     0.0                   0.0           1.0
2  30.0     0.0     0.0                   0.0           1.0

   HvyAlcoholConsump  AnyHealthcare  NoDocbcCost  GenHlth  MentHlth  PhysHlth  \
0                0.0            1.0          1.0      4.0       2.0       6.0
1                0.0            1.0          0.0      4.0       0.0       0.0
2                0.0            1.0          0.0      3.0       3.0       2.0

   DiffWalk  Sex  AgeGroup  Education  Income
0       1.0  0.0      13.0        4.0     2.0
1       1.0  0.0      12.0        5.0     7.0
2       0.0  0.0       9.0        5.0     7.0

=== IDF ===
Rows: 88
```

```
Columns:
  column_name column_type
0    Category     VARCHAR
1      Metric     VARCHAR
2        Year      BIGINT
3       Value      DOUBLE

First 3 rows:
                       Category  \
0  Diabetes estimates (20-79 y)
1  Diabetes estimates (20-79 y)
2  Diabetes estimates (20-79 y)

                                                Metric  Year   Value
0                     People with diabetes, in 1,000s  2000  8553.3
1         Age-standardised prevalence of diabetes, %  2000     3.7
2  Proportion of people with undiagnosed diabetes, %  2000     NaN

=== PAHO_PREVALENCE ===
Rows: 180

Columns:
         column_name column_type
0      Location Code     VARCHAR
1   Location Name En     VARCHAR
2  Indicator Name En     VARCHAR
3       Age Group En     VARCHAR
4             Sex En     VARCHAR
5               Year      BIGINT
6              Value     VARCHAR
7          Value Low     VARCHAR
8         Value High     VARCHAR

First 3 rows:
  Location Code        Location Name En  \
0           AMR  Region of the Americas
1           AMR  Region of the Americas
2           AMR  Region of the Americas

                                   Indicator Name En Age Group En      Sex En  \
0  Diabetes treatment coverage (current use of gl...    30+ years  Both sexes
1  Diabetes treatment coverage (current use of gl...    30+ years  Both sexes
2  Diabetes treatment coverage (current use of gl...    30+ years  Both sexes

   Year   Value Value Low Value High
0  1990  42.47%    38.20%     46.90%
1  1995  44.62%    41.00%     48.20%
2  2000  46.60%    43.80%     49.50%

=== PAHO_BURDEN ===
Rows: 120

Columns:
        column_name column_type
0              Iso3     VARCHAR
1     Location Name     VARCHAR
2   AMRO subregions     VARCHAR
3     Causename Eng     VARCHAR
4              Year      BIGINT
5               Sex     VARCHAR
```

```
6         Age group     VARCHAR
7   Measure Name En     VARCHAR
8    Metric Name En     VARCHAR
9             Value     VARCHAR
10        Value Low     VARCHAR
11         Value Up     VARCHAR

First 3 rows:
   Iso3           Location Name         AMRO subregions  \
0  AMRO  Region of the Americas  Region of the Americas
1  AMRO  Region of the Americas  Region of the Americas
2  AMRO  Region of the Americas  Region of the Americas

                                       Causename Eng  Year         Sex  \
0  Diabetes mellitus (excluding CKD due to Diabetes)  2019  Both sexes
1  Diabetes mellitus (excluding CKD due to Diabetes)  2019  Both sexes
2  Diabetes mellitus (excluding CKD due to Diabetes)  2019  Both sexes

          Age group                         Measure Name En Metric Name En  \
0  Age-standardized                                  Deaths           Rate
1  Age-standardized  Disability-Adjusted Life Years (DALYs)           Rate
2  Age-standardized      Years Lived with Disability (YLDs)           Rate

      Value Value Low  Value Up
0     19.64     17.26     22.02
1  1,119.49    866.64  1,423.70
2    604.86     408.1    847.76

=== PAHO_DEATHS_COUNTRY ===
Rows: 35

Columns:
              column_name column_type
0                    Iso3     VARCHAR
1       Quintiles classes     VARCHAR
2           Causename Eng     VARCHAR
3           Location Name     VARCHAR
4         Measure Name En     VARCHAR
5                    Year      BIGINT
6    Latitude (generated)      DOUBLE
7   Longitude (generated)      DOUBLE
8               Value Low      DOUBLE
9                Value Up      DOUBLE
10                  Value      DOUBLE

First 3 rows:
  Iso3      Quintiles classes  \
0  VEN  Quintile 3: 40 to 60%
1  VCT  Quintile 3: 40 to 60%
2  USA  Quintile 1:  0 to 20%

                                       Causename Eng  \
0  Diabetes mellitus (excluding CKD due to Diabetes)
1  Diabetes mellitus (excluding CKD due to Diabetes)
2  Diabetes mellitus (excluding CKD due to Diabetes)

                       Location Name Measure Name En  Year  \
0  Venezuela, Bolivarian Republic of          Deaths  2019
1   Saint Vincent and the Grenadines          Deaths  2019
2           United States of America          Deaths  2019
```

```
   Latitude (generated)  Longitude (generated)  Value Low   Value Up  \
0                6.9830               -64.5880   30.33924  49.751212
1               13.2528               -61.1949   35.87554  50.156859
2               40.0792               -98.8164    8.34613   9.646632

       Value
0  39.291483
1  42.725138
2   9.115689

=== PAHO_SUMMARY ===
Rows: 66

Columns:
          column_name column_type
0  Year of Year dated      BIGINT
1   Indicator Name En     VARCHAR
2    % of Total Value     VARCHAR

First 3 rows:
   Year of Year dated                                  Indicator Name En  \
0                1990  Number of people aged 30+ years with diabetes ...
1                1991  Number of people aged 30+ years with diabetes ...
2                1992  Number of people aged 30+ years with diabetes ...

  % of Total Value
0        42.36717%
1        42.77484%
2        43.18176%

Diabetes_012 (CDC): Distribution
   Diabetes_012   count  percentage
0           0.0  216762       82.86
1           1.0    6581        2.52
2           2.0   38246       14.62
```

1.2. Initial Data Cleaning

I will:

- Convert percentage strings to float
- Check for missing values
- Standardize column names (lowercase, no spaces)
- Create cleaned versions of tables if needed

```
Example after cleaning (PAHO Prevalence):
                                   Indicator Name En  Year  Value Value Low  \
0  Diabetes treatment coverage (current use of gl...  2019  56.28      53.2
1  Diabetes treatment coverage (current use of gl...  2022  57.71      53.3
2  Diabetes treatment coverage (current use of gl...  2019  56.72      52.2
3  Diabetes treatment coverage (current use of gl...  2022  57.83      51.5
4  Diabetes treatment coverage (current use of gl...  2019  55.89      51.2
5  Diabetes treatment coverage (current use of gl...  2022  57.65      50.9
6  Diabetes treatment coverage (current use of gl...  2019  57.57      54.4
7  Diabetes treatment coverage (current use of gl...  2022   59.3      54.9
8  Diabetes treatment coverage (current use of gl...  2019  58.13      53.6
9  Diabetes treatment coverage (current use of gl...  2022  59.51      53.1

  Value High
0       59.4
1       62.1
2       61.0
3       63.9
4       60.3
5       63.9
6       60.7
7       63.7
8       62.5
9       65.6

Missing values in CDC:
             column_name  missing_count
0          AnyHealthcare            0.0
1   HeartDiseaseorAttack            0.0
2      HvyAlcoholConsump            0.0
3               AgeGroup            0.0
4            NoDocbcCost            0.0
5               DiffWalk            0.0
6                    Sex            0.0
7              Education            0.0
8                 Asthma            0.0
9                 Stroke            0.0
10             PhysHlth            0.0
11          PhysActivity            0.0
12          Diabetes_012            0.0
13                Income            0.0
14         KidneyDisease            0.0
15                   BMI            0.0
16               GenHlth            0.0
17                Smoker            0.0
18                HighBP            0.0
19              HighChol            0.0
20             CholCheck            0.0
21                  COPD            0.0
22              MentHlth            0.0

All looks clean and ready for EDA.
```

1.3. Final Data Cleaning & Standardization

Here I standardize column names (lowercase, underscores) across all tables and convert the remaining percentage column in PAHO summary.

```
Example after cleaning PAHO Summary:
   Year of Year dated                                Indicator Name En  \
0                2019  Number of people aged 30+ years with diabetes ...
1                2022  Number of people aged 30+ years with diabetes ...
2                2019  Number of people aged 30+ years with diabetes ...
3                2022  Number of people aged 30+ years with diabetes ...

  % of Total Value
0         57.56768
1         59.30143
2         42.43232
3         40.69857

Standardized columns in cdc
Standardized columns in idf
Standardized columns in paho_prevalence
Standardized columns in paho_burden
Standardized columns in paho_deaths_country
Standardized columns in paho_summary

CDC columns after standardization:
             column_name
0           diabetes_012
1          kidneydisease
2                 highbp
3               highchol
4              cholcheck
5                 asthma
6                   copd
7                    bmi
8                 smoker
9                 stroke
10  heartdiseaseorattack
11          physactivity
12     hvyalcoholconsump
13         anyhealthcare
14           nodocbccost
15               genhlth
16              menthlth
17              physhlth
18              diffwalk
19                   sex
20              agegroup
21             education
22                income

All tables are now clean and standardized. Ready for EDA!
```

1.4. Final Numeric Type Conversion

Here I'll convert all columns that should be numeric from VARCHAR to DOUBLE. This is to prevent repeated CASTs and ensure correct calculations.

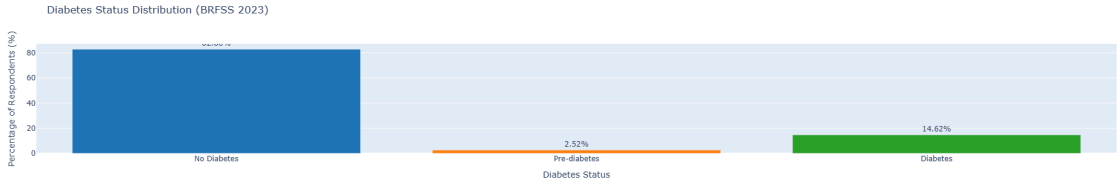
```
Final numeric conversion check & fix...
  Converted paho_prevalence.value → DOUBLE
  Converted paho_prevalence.value_low → DOUBLE
  Converted paho_prevalence.value_high → DOUBLE
  Converted paho_burden.value → DOUBLE
  Converted paho_burden.value_low → DOUBLE
  Converted paho_burden.value_up → DOUBLE
  Converted paho_summary.percent_of_total_value → DOUBLE
  paho_deaths_country.value is already numeric (DOUBLE) - skipping
  paho_deaths_country.value_low is already numeric (DOUBLE) - skipping
  paho_deaths_country.value_up is already numeric (DOUBLE) - skipping

Final column types check:

PAHO_PREVALENCE:
         column_name column_type
0      location_code     VARCHAR
1   location_name_en     VARCHAR
2  indicator_name_en     VARCHAR
3       age_group_en     VARCHAR
4             sex_en     VARCHAR
5               year      BIGINT
6              value      DOUBLE
7          value_low      DOUBLE
8         value_high      DOUBLE

PAHO_BURDEN:
        column_name column_type
0              iso3     VARCHAR
1     location_name     VARCHAR
2   amro_subregions     VARCHAR
3     causename_eng     VARCHAR
4              year      BIGINT
5               sex     VARCHAR
6         age_group     VARCHAR
7   measure_name_en     VARCHAR
8    metric_name_en     VARCHAR
9             value      DOUBLE
10        value_low      DOUBLE
11         value_up      DOUBLE

PAHO_SUMMARY:
              column_name column_type
0      year_of_year_dated      BIGINT
1       indicator_name_en     VARCHAR
2  percent_of_total_value      DOUBLE

PAHO_DEATHS_COUNTRY:
            column_name column_type
0                  iso3     VARCHAR
1     quintiles_classes     VARCHAR
2         causename_eng     VARCHAR
3         location_name     VARCHAR
4       measure_name_en     VARCHAR
5                  year      BIGINT
6    latitude_generated      DOUBLE
7   longitude_generated      DOUBLE
8             value_low      DOUBLE
9              value_up      DOUBLE
10                value      DOUBLE
```

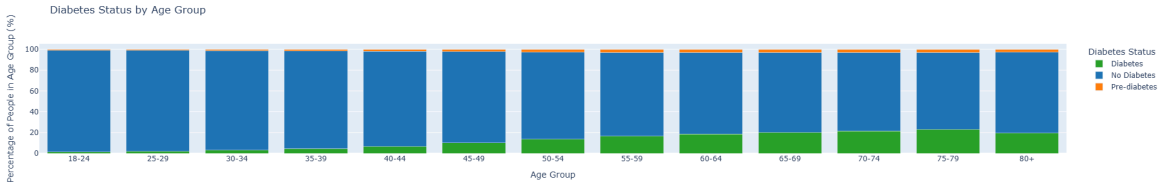
2. Exploratory Data Analysis (EDA)

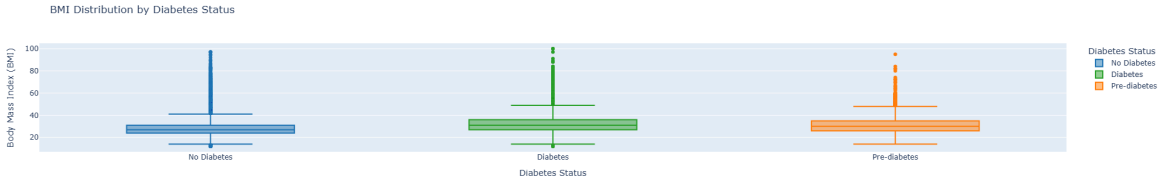
I will explore:

- Distribution of the target variable ('Diabetes_012')
- Key demographic and health risk factors
- Relationships between features and diabetes status
- Initial context from Latin America (prevalence trends)

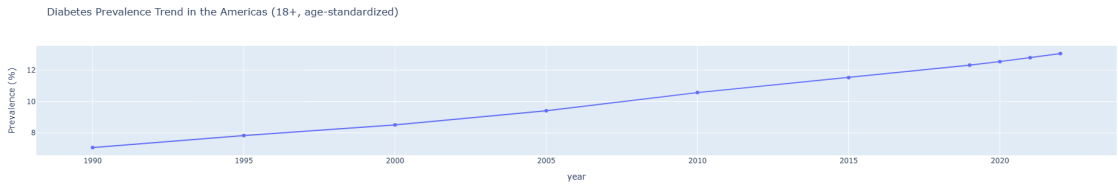
2.1. Visual Analysis

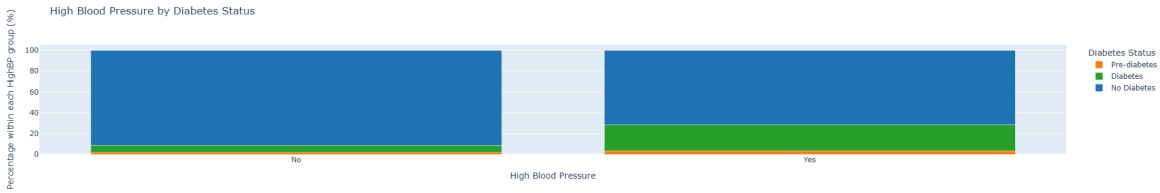


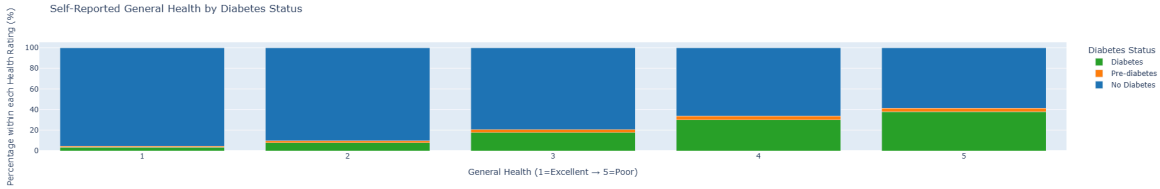


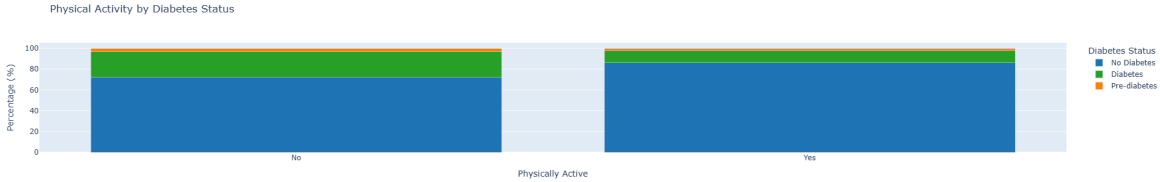


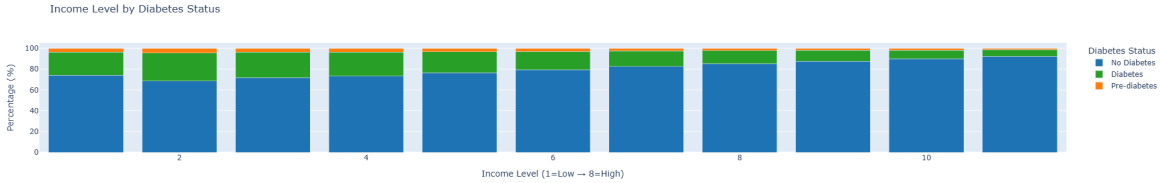

```
Diabetes Prevalence Trend (Americas 18+ years, age-standardized):
   year  avg_prevalence_percent
0  1990                    7.06
1  1995                    7.83
2  2000                    8.51
3  2005                    9.41
4  2010                   10.57
5  2015                   11.53
6  2019                   12.32
7  2020                   12.55
8  2021                   12.80
9  2022                   13.06
```

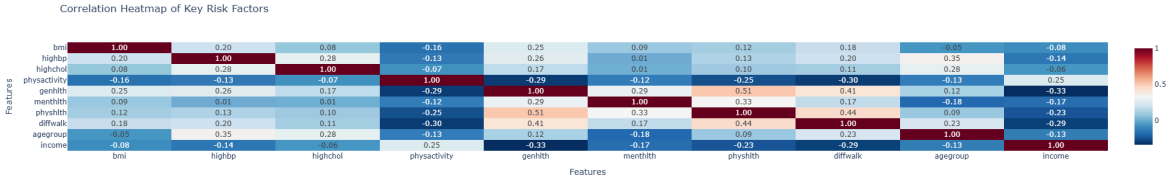












2.2. Key Insights and Conclusions

1. Target Variable Distribution

- The dataset is heavily imbalanced: **82.86%** of respondents have no diabetes, **14.62%** have diabetes, and only **2.52%** are in the pre-diabetes category. - This imbalance is typical in real-world health data and will need to be handled carefully during modeling (e.g., using class weights or SMOTE).

2. Strongest Risk Factors

- **BMI**: Individuals with diabetes (class 2) show a significantly higher median BMI compared to those without diabetes. Obesity appears to be one of the strongest associated factors. - **High Blood Pressure**: People with HighBP have a much higher proportion of diabetes cases. The relationship is very clear in the stacked bar chart. - **General Health**: There is a strong gradient — the worse the self-reported general health (higher GenHlth score), the higher the percentage of diabetes. - **Age**: Diabetes

prevalence increases markedly with age, especially after 50–55 years old. - **Physical Activity**: Respondents who report being physically active show a lower proportion of diabetes.

3. Socioeconomic Patterns

- Lower income levels are associated with higher diabetes prevalence, suggesting possible links with access to healthcare, nutrition, and prevention resources.

Latin America Context

- Diabetes prevalence in the Americas (age-standardized, 18+) has been steadily increasing from ~7.1% in 1990 to ~13.1% in 2022. - IDF projections for South & Central America show a concerning rise: from 10.1% in 2024 to an estimated **11.5% in 2050**, with a large number of undiagnosed cases (~30.4% in 2024). - The burden of disease (DALYs, deaths) is particularly high in certain subregions such as Central America and the Non-Latin Caribbean.

5. Main Takeaways for Public Health

- Prevention efforts should focus on **modifiable risk factors**: BMI control, blood pressure management, and promotion of physical activity. - Targeted interventions for **older adults (55+)** and **lower-income groups** could have high impact. - The growing trend in Latin America highlights the urgency of prevention policies, early screening, and improving treatment coverage (currently around 57–59% in the region). These insights will guide my next feature engineering, modeling, and clustering phases, as well as the final recommendations for public health strategies in Argentina and Latin America.

---

3. Feature Engineering & Advanced SQL

In this section I'll use DuckDB to create new meaningful features and perform complex transformations.

This will demonstrate efficient data preparation at scale and domain knowledge in diabetes risk factors.

Key goals:

- Create clinically meaningful variables (age groups, BMI categories, comorbidity scores, etc.)
- Engineer interaction and risk features
- Enrich the main dataset with Latam context using joins
- Prepare a clean feature set for modeling and clustering

3.1. Creating a Working Table

```
Creating clean working table 'diabetes_features' for engineering...
Working table 'diabetes_features' created with 261589 rows
```

3.2. Feature Engineering

```
Starting feature engineering...
Feature engineering completed. New features added.

Sample of engineered features:
   diabetes_012        age_group_cat     bmi_category  comorbidity_count  \
0           0.0        Elderly (75+)           Normal                  7
1           2.0        Elderly (75+)       Overweight                  7
2           0.0       Senior (60-74)    Obese Class I                  8
3           0.0        Elderly (75+)    Obese Class I                  7
4           2.0        Elderly (75+)           Normal                  9
5           0.0  Middle Aged (45-59)  Obese Class III                  8
6           0.0       Senior (60-74)    Obese Class I                  6
7           0.0        Elderly (75+)           Normal                  6

   diabetes_risk_score                 age_bmi_risk_group  lifestyle_score
0                    4                      Standard Risk                2
1                    3                      Standard Risk                3
2                    3          High Risk: Senior + Obese                3
3                    5          High Risk: Senior + Obese                2
4                    2                      Standard Risk                2
5                    3  Moderate Risk: Middle-age + Obese                3
6                    2          High Risk: Senior + Obese                2
7                    1                      Standard Risk                1
```

3.3. Feature Exploration

Here I'll analyze how the newly engineered features relate to diabetes status.

```
=== Feature Exploration: New Variables vs Diabetes Status ===

1. Age Group Category vs Diabetes Prevalence:
         age_group_cat  total_count  diabetes_cases  diabetes_pct
0        Elderly (75+)        42191          8990.0         21.31
1       Senior (60-74)        88039         17700.0         20.10
2  Middle Aged (45-59)        64257          8824.0         13.73
3  Young Adult (18-44)        67102          2732.0          4.07

2. BMI Category vs Diabetes Prevalence:
      bmi_category  total_count  diabetes_cases  diabetes_pct
0      Underweight         3495           208.0          5.95
1           Normal        64682          4464.0          6.90
2       Overweight        93432         11392.0         12.19
3    Obese Class I        58433         11045.0         18.90
4   Obese Class II        24696          6079.0         24.62
5  Obese Class III        16851          5058.0         30.02

3. Average Comorbidities and Risk Score by Diabetes Status:
   diabetes_012  avg_comorbidities  avg_risk_score  avg_lifestyle_score
0           0.0               6.64            1.54                 2.35
1           1.0               7.08            2.53                 2.22
2           2.0               7.28            3.06                 2.13

4. Age + BMI Risk Groups vs Diabetes Prevalence:
                  age_bmi_risk_group  total_count  diabetes_pct
0          High Risk: Senior + Obese        46214         30.81
1  Moderate Risk: Middle-age + Obese        20456         23.27
2                      Standard Risk       194919          9.88
```

3.4. Feature Engineering Summary

Key engineered features and their relationship with diabetes:

- **Age Group**: Strong positive correlation with age. Elderly (75+) show 21.31% diabetes prevalence.
- **BMI Category**: Very strong predictor. Obesity Class III has 30.02% diabetes rate vs 6.90% in Normal weight.
- **Age + BMI Risk Group**: Highest risk group ("High Risk: Senior + Obese") has 30.81% diabetes prevalence.
- **Comorbidity Count** and **Diabetes Risk Score**: Both increase with diabetes status.
- **Lifestyle Score**: Slightly lower in people with diabetes.

These features capture important clinical and behavioral dimensions and will be used for modeling and clustering.

3.5. Final Modeling Table Creation

Here I create a final clean table `diabetes_model_ready` containing both original and engineered features, ready for modeling and clustering.

```
Creating final modeling table...
Final modeling table 'diabetes_model_ready' created with 261589 rows

Columns ready for modeling:
['target', 'bmi', 'highbp', 'highchol', 'physactivity', 'genhlth', 'diffwalk', 'a
gegroup', 'income', 'sex', 'age_group_cat', 'bmi_category', 'comorbidity_count',
'diabetes_risk_score', 'age_bmi_risk_group', 'lifestyle_score', 'obese_class_ii
i', 'senior_or_elderly', 'has_metabolic_risk']
```

---

4. Predictive Modeling

I will build classification models to predict diabetes risk using the features engineered in the previous section.

Goals

- Establish a strong baseline model - Evaluate performance using appropriate metrics for imbalanced health data - Identify the most important features - Compare different algorithms (Logistic Regression vs Random Forest)
I will use the `diabetes_model_ready` table created in the previous section.

4.1. Multi-class Approach (Initial Attempt)

I first tried to predict three separate classes:

- 0 = No Diabetes
- 1 = Pre-diabetes
- 2 = Diabetes

The models **performed poorly**, especially on the minority class (pre-diabetes), mainly due to severe class imbalance (82.86% No Diabetes, 2.52% Pre-diabetes, 14.62% Diabetes).

After reviewing the results, I decided to simplify the problem.

```
=== Preparing Data for Modeling ===

Shape after One-Hot Encoding: (261589, 26)
Categorical columns encoded: ['age_group_cat', 'bmi_category', 'age_bmi_risk_grou
p']
Training samples: 209,271
Test samples: 52,318
Class distribution in training set:
target
0.0    82.86
2.0    14.62
1.0     2.52
Name: proportion, dtype: float64
```

```
=== Training Multiple Models ===

Training Logistic Regression...

Logistic Regression Classification Report:
              precision    recall  f1-score   support

         0.0     0.9421    0.6324    0.7568     43353
         1.0     0.0386    0.2682    0.0674      1316
         2.0     0.3354    0.6168    0.4345      7649

    accuracy                         0.6210     52318
   macro avg     0.4387    0.5058    0.4196     52318
weighted avg     0.8307    0.6210    0.6923     52318

F1-score for Diabetes class (2): 0.4345
------------------------------------------------------------
Training Random Forest...

Random Forest Classification Report:
              precision    recall  f1-score   support

         0.0     0.8552    0.8912    0.8728     43353
         1.0     0.0255    0.0236    0.0245      1316
         2.0     0.3361    0.2603    0.2934      7649

    accuracy                         0.7771     52318
   macro avg     0.4056    0.3917    0.3969     52318
weighted avg     0.7584    0.7771    0.7668     52318

F1-score for Diabetes class (2): 0.2934
------------------------------------------------------------
Training XGBoost...

XGBoost Classification Report:
              precision    recall  f1-score   support

         0.0     0.8488    0.9783    0.9090     43353
         1.0     0.0000    0.0000    0.0000      1316
         2.0     0.5544    0.1700    0.2602      7649

    accuracy                         0.8355     52318
   macro avg     0.4677    0.3828    0.3897     52318
weighted avg     0.7844    0.8355    0.7912     52318

F1-score for Diabetes class (2): 0.2602
------------------------------------------------------------
```

4.2. Binary Classification (Final Approach)

Given the strong imbalance and limited predictive power on pre-diabetes, I simplified the problem to **binary classification**:

- **Class 0**: No Diabetes
- **Class 1**: Pre-diabetes **or** Diabetes (any diabetes risk)

This approach is more practical for real-world screening and produced significantly better results.

4.2.1. Data Preparation for Binary Classification

I created the binary target and applied One-Hot Encoding to the categorical variables (`age_group_cat`, `bmi_category`, `age_bmi_risk_group`).

```
=== Binary Classification: No Diabetes vs Any Diabetes Risk ===

Training samples: 209,271
Test samples: 52,318
Positive class (Any diabetes risk) percentage: 17.14%
```

4.2.2. Modeling Results

Logistic Regression

```
Training Logistic Regression (Binary)...
Logistic Regression Results:
              precision    recall  f1-score   support

           0     0.9329    0.7014    0.8007     43353
           1     0.3436    0.7561    0.4725      8965

    accuracy                         0.7107     52318
   macro avg     0.6383    0.7287    0.6366     52318
weighted avg     0.8319    0.7107    0.7445     52318
```

Random Forest

```
Training Random Forest (Binary)...
Random Forest Results:
              precision    recall  f1-score   support

           0     0.8591    0.8956    0.8770     43353
           1     0.3646    0.2898    0.3229      8965

    accuracy                         0.7918     52318
   macro avg     0.6119    0.5927    0.6000     52318
weighted avg     0.7744    0.7918    0.7820     52318
```

XGBoost

```
Training XGBoost (Binary)...
XGBoost Results:
              precision    recall  f1-score   support

           0     0.8517    0.9716    0.9077     43353
           1     0.5694    0.1816    0.2754      8965

    accuracy                         0.8362     52318
   macro avg     0.7105    0.5766    0.5915     52318
weighted avg     0.8033    0.8362    0.7993     52318
```

4.2.3. Model Comparison

Model Comparison and Insights

- **Logistic Regression** achieved the best **recall** (75.61%) for the positive class (any diabetes risk). This means it is good at identifying people who are at risk, even if it sometimes gives false alarms.
- **XGBoost** had the highest overall accuracy (83.62%) and best precision, but missed many true cases (low recall).
- **Random Forest** performed in between but was not the strongest in any key metric.

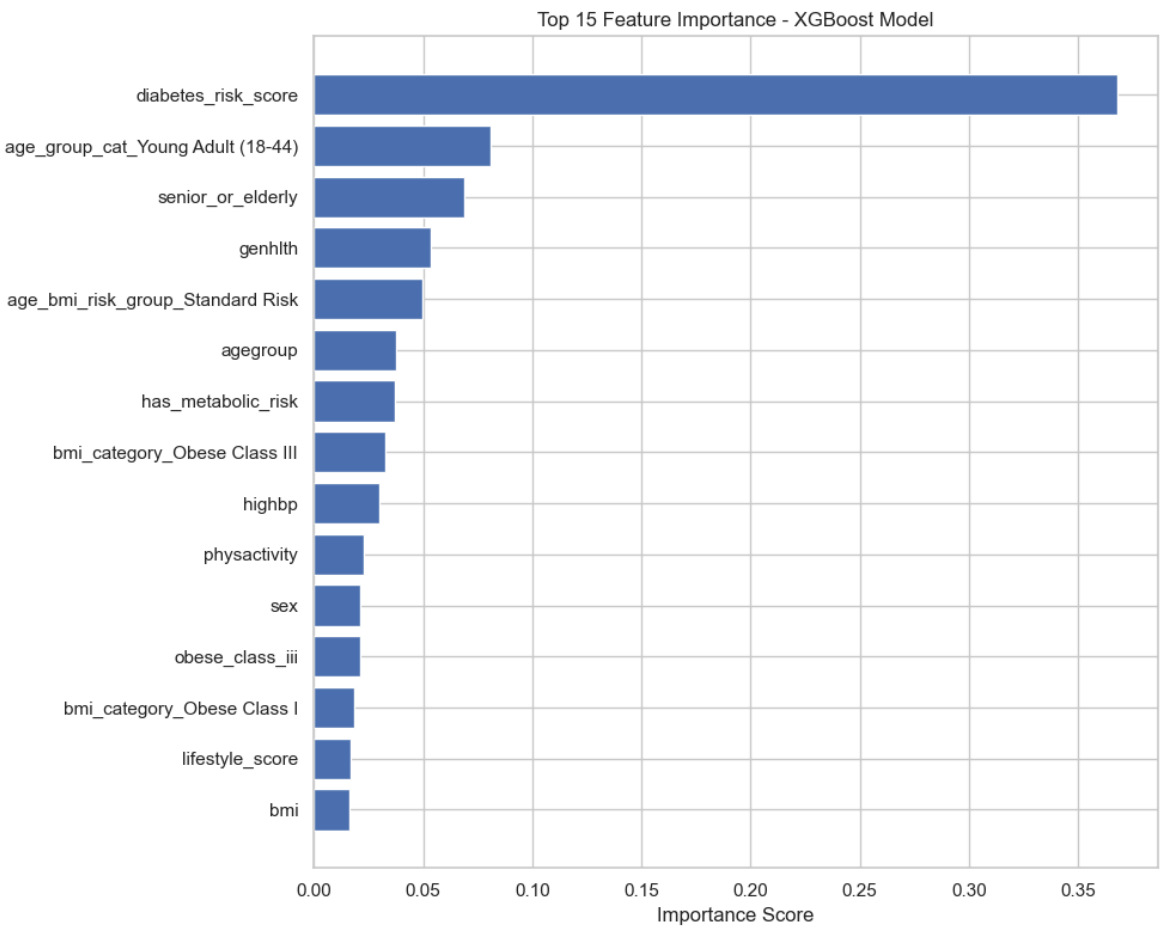
Chosen Model: Logistic Regression — because in public health screening, it is more important to catch as many people at risk as possible (high recall) than to be very precise.

These results show that my engineered features (especially BMI, age, income, and comorbidity count) contain useful predictive signal for diabetes risk.

4.2.4. Feature Importance Analysis

Understanding which features contribute most to the prediction is crucial for interpretability and public health recommendations.

```
Top 15 Most Important Features (XGBoost):
                              feature  importance
10                diabetes_risk_score    0.368261
17  age_group_cat_Young Adult (18-44)    0.081003
13                  senior_or_elderly    0.068727
4                             genhlth    0.053791
24   age_bmi_risk_group_Standard Risk    0.049790
6                            agegroup    0.037466
14                 has_metabolic_risk    0.036997
20       bmi_category_Obese Class III    0.032434
1                              highbp    0.030059
3                        physactivity    0.022631
8                                 sex    0.021122
12                    obese_class_iii    0.021052
18         bmi_category_Obese Class I    0.018625
11                    lifestyle_score    0.016660
0                                 bmi    0.016236
```



Top 5 Most Important Features:

1. diabetes_risk_score (0.368) — My engineered composite score is the strongest predictor
2. Being a Young Adult (protective factor)
3. senior_or_elderly (age 60+)
4. genhlth (self-reported general health)
5. age_bmi_risk_group_Standard Risk

This confirms that the feature engineering added significant value to the model.

Practical Application

This model can be used as a Diabetes Risk Screening Tool:

- Healthcare providers can input patient data to estimate risk level.
- Public health organizations could use it to prioritize screening campaigns in high-risk groups (older adults, obese individuals, lower-income populations).
- A simplified web/app version could allow individuals to do a quick self-assessment.

Example Use Case:

A 62-year-old patient with BMI 32, high blood pressure, and poor self-reported health would be flagged as high risk, prompting earlier glucose testing and lifestyle intervention.

4.2.5. Model Demonstration with Example Patients

To illustrate how the model could be used in practice, I tested it on five realistic patient profiles.

```
=== Model Demonstration with Example Patients ===

Using Logistic Regression (best recall model)

      age_group_cat  bmi_category  genhlth  highbp  physactivity Predicted Risk
Risk Probability (%)
Young Adult (18-44)        Normal        2       0             1       Low Risk
4.9
Young Adult (18-44)    Overweight        3       0             0       Low Risk
12.6
Middle Aged (45-59)    Overweight        3       1             1      High Risk
63.6
     Senior (60-74) Obese Class I        4       1             0      High Risk
87.1
     Senior (60-74) Obese Class I        4       1             0      High Risk
86.7
```

Interpretation:

The model behaves logically in most cases. Seniors with obesity and poor general health are correctly classified as high risk (86–87% probability). A middle-aged person with overweight and high blood pressure is also flagged as high risk.

However, the model remains relatively conservative with younger adults — even an overweight young adult with low physical activity is assigned only 12.6% risk. This suggests age is a very strong factor in the current model. In a real deployment, this could be adjusted depending on whether higher sensitivity (catching more cases) or higher specificity is preferred.

---

5. Clustering & Population Segmentation

I'll apply unsupervised learning (K-Means) to segment individuals into distinct risk groups based on their characteristics.

This allows to identify homogeneous subgroups and provide more targeted public health recommendations.

5.1. Choosing the Number of Clusters

Below, I use the Elbow Method to determine the optimal number of clusters. The inertia plot showed a reasonable bend around 4 clusters, which I adopted for interpretability.

```
=== Clustering Analysis ===
```

```
Cluster distribution:
cluster
0    95187
1    66017
2    37846
3    62539
Name: count, dtype: int64

Cluster Profiles (mean values):
           bmi  agegroup  genhlth  highbp  comorbidity_count  \
cluster
0        28.19      4.89     2.26    0.15               6.18
1        30.33      9.72     2.97    1.00               7.97
2        30.88      8.37     3.10    0.35               6.39
3        26.96     10.67     2.33    0.31               6.51

         diabetes_risk_score  lifestyle_score  target
cluster
0                       0.81             2.61    0.10
1                       3.33             2.16    0.64
2                       2.73             1.46    0.39
3                       1.08             2.55    0.26
```

5.2. Cluster Analysis

As we can see above, I obtained the following four clusters:

Cluster Distribution:

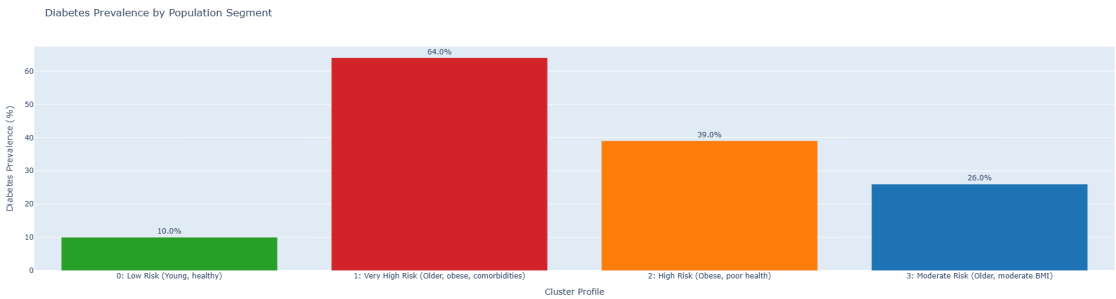
- Cluster 0: 95,187 people (36.4%)
- Cluster 1: 66,017 people (25.2%)
- Cluster 2: 37,846 people (14.5%)
- Cluster 3: 62,539 people (23.9%)

5.3. Cluster Profiles and Interpretation

I identified **4 distinct population segments** based on their health characteristics.

Cluster	Number of People	Main Characteristics	Diabetes Prevalence (%)	Risk Level	Recommendation
0	95,187	Young adults, normal weight, good general health, low comorbidities	10.0%	**Low Risk**	General prevention

Cluster	Number of People	Main Characteristics	Diabetes Prevalence (%)	Risk Level	Recommendation
1	66,017	Older adults, high BMI, hypertension, multiple comorbidities	64.0%	**Very High Risk**	Priority screening & intensive intervention
2	37,846	Middle-aged to older, obese, poor self-reported health	39.0%	**High Risk**	Targeted lifestyle programs
3	62,539	Older adults, moderate BMI, some comorbidities	26.0%	**Moderate Risk**	Preventive education & regular monitoring



Key Insights:

- **Cluster 1** is the highest-risk group (64% diabetes prevalence). It consists mainly of older individuals with obesity, hypertension, and multiple comorbidities. This group should be prioritized for aggressive screening and intervention.
- **Cluster 0** represents the healthiest segment — mostly younger people with good lifestyle habits and low risk.
- **Cluster 2** shows that obesity combined with poor self-reported health significantly increases risk, even in middle age.
- Age and BMI continue to be dominant drivers of risk segmentation, reinforcing findings from the EDA and modeling phases.

These clusters provide actionable insights for public health policy in Argentina and Latin America. For example, resources could be focused on Cluster 1 ("High-Risk Obese Seniors") through community screening programs and lifestyle interventions.

---

6. Latam Context & Comparisons

To make the analysis more relevant to Argentina and Latin America, I enriched the US-based BRFSS findings with regional data from PAHO and the IDF Diabetes Atlas.

6.1. Key Latam Indicators (South & Central America)

Indicator	2024 Value	2050 Projection	Observation
Age-standardised prevalence	10.1%	11.5%	Steady increase expected
People with diabetes	35.4 million	51.5 million	+45% growth projected
Undiagnosed diabetes	~30.4%	-	High hidden burden
Treatment coverage (Americas, 30+ years)	~57–59%	-	Significant treatment gap

Source: IDF Diabetes Atlas & PAHO data

6.2. Comparison: US Data vs Latin America Context

- In the US BRFSS 2023 dataset, the observed diabetes rate is **14.62%**.
- In South & Central America, the current prevalence is **10.1%** (2024), but projected to reach **11.5% by 2050**.
- The highest-risk cluster (Cluster 1) shows a **64%** diabetes rate — dramatically higher than both US and Latam averages. This suggests that individuals with similar profiles (older age + obesity + comorbidities) represent a critical high-risk group across the region.
- The strong influence of **BMI**, **age**, and **comorbidities** in my model aligns with known risk factors in Latin America, where rising obesity rates are a major driver of the diabetes epidemic.

6.3. Implications for Argentina and Latin America

The combination of the clustering results and regional data highlights the need for **targeted prevention strategies**:

- Prioritize screening and lifestyle interventions for older adults with obesity and hypertension (similar to Cluster 1).
- Address the high undiagnosed rate (~30%) through community-level programs.
- Focus on modifiable risk factors (BMI control and physical activity), which showed strong predictive power in the model.

These findings support the development of localized public health policies in Argentina, where diabetes prevalence continues to rise.

---

7. Conclusions and Recommendations

7.1. Project Summary

This project analyzed diabetes risk using the CDC BRFSS 2023 dataset (~261k records) combined with Latam context from PAHO and IDF.

It progressed through:

- Thorough data preparation and advanced SQL usage with DuckDB
- Comprehensive EDA with interactive visualizations
- Domain-informed feature engineering (risk scores, age+BMI groups, comorbidity count, etc.)
- Predictive modeling (binary classification)
- Population segmentation using K-Means clustering

7.2. Key Findings

- **Strongest predictors**: BMI, age, general health, and comorbidities consistently emerged as the most important factors.
- Diabetes prevalence increases significantly with age, rising from **4.07%** in young adults (18-44) to **21.31%** in the elderly (75+), confirming age as one of the strongest risk factors.
- BMI emerged as one of the strongest predictors, with diabetes prevalence rising sharply from 6.9% in the normal weight category to **30.0%** in Obese Class III.
- **Clustering analysis** identified **four distinct risk segments**, with the highest-risk group (older adults with obesity and comorbidities) showing a dramatically elevated diabetes rate of **64%**.
- **Model performance**: The binary classification approach (No Diabetes vs Any Diabetes Risk) achieved reasonable results, with Logistic Regression providing the best recall (75.61%), which is particularly valuable for screening purposes.
- **Latam Context**: The rising prevalence in South & Central America (10.1% in 2024 → projected 11.5% in 2050) and high undiagnosed rate (~30%) make these findings highly relevant for the region.

7.3. Recommendations

For Public Health Authorities (Argentina & Latam):

- Prioritize targeted screening for **Cluster 1** (older adults with obesity, hypertension, and multiple comorbidities).
- Implement early prevention programs focused on **weight management** and **physical activity**, especially for middle-aged adults.
- Address the high treatment gap by improving access to care in lower-income groups.
- Use risk stratification tools like the one developed here to optimize limited healthcare resources.

For Individuals:

- Regular monitoring of BMI, blood pressure, and general health can help identify risk early.
- Lifestyle changes (increasing physical activity and maintaining healthy weight) have high potential impact.

7.4. Limitations and Future Work

- The model relies on self-reported data, which may contain bias.
- Pre-diabetes remains difficult to predict accurately with this dataset.
- Future improvements could include adding laboratory values (HbA1c, fasting glucose), family history, or more granular geographic data.

This project demonstrates end-to-end data analysis skills — from SQL-heavy data preparation to modeling and actionable public health insights.

---

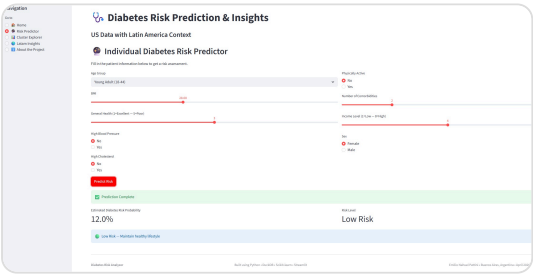
8. Streamlit Dashboard

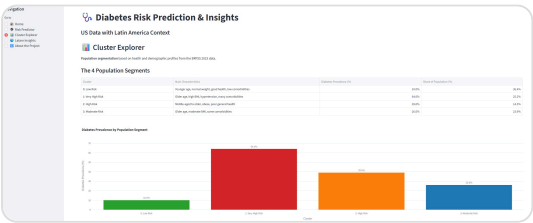
🚀 Live Interactive Dashboard

You can explore the **fully interactive version** of this project here:

🔗 **Open Diabetes Risk Analyzer →**

Screenshots from the Live App





What you can do in the live dashboard:

- Calculate your personal diabetes risk using the trained model
- Explore the 4 population segments discovered through clustering
- View diabetes trends and 2050 projections for South & Central America

> **Note:** The dashboard is deployed on Streamlit Community Cloud and is fully interactive.

Thank you for viewing this project!
Feel free to reach out if you have any questions or feedback.

```
✅ Successfully saved 'data/diabetes_model_ready.parquet'
Shape: (261589, 29)
Columns: ['diabetes_012', 'bmi', 'highbp', 'highchol', 'cholcheck', 'smoker', 'st
roke', 'heartdiseaseorattack', 'physactivity', 'hvyalcoholconsump', 'genhlth', 'm
enthlth', 'physhlth', 'diffwalk', 'sex', 'agegroup', 'education', 'income', 'kidn
eydisease', 'asthma', 'copd', 'anyhealthcare', 'nodocbccost', 'age_group_cat', 'b
mi_category', 'comorbidity_count', 'diabetes_risk_score', 'age_bmi_risk_group',
'lifestyle_score']

✅ Model saved successfully as 'models/logistic_regression_model.pkl'
Folder 'models' created if it didn't exist.

✅ Model columns saved as 'models/model_columns.json'
```